



AgroLnk — Complete Production Demo & Testing Manual

A Step-by-Step Interactive Guide for QA Testers & Stakeholder Demonstrations

Executive Overview

AgroLnk is an end-to-end B2B Agri-Trade Platform integrating **Spot Commerce, Live Reverse-English Auctions, 100% Escrow & 0.50% Commission, Multi-Axle Logistics, WDRA e-NWR Warehouses, 30-Day Working Capital Financing, Assayer Quality Inspections, and Anti-Circumvention Privacy Chat.**

This manual guides you step-by-step through every user role and business flow with exact account credentials, navigation paths, actions to perform, and expected results.

1. Test Accounts & Role Access Matrix

USER ROLE	PERSONA NAME	LOGIN EMAIL	PASSWORD	PRIMARY PORTAL ROUTE	KEY TESTING RESPONSIBILITIES
Platform Admin	Platform Admin	admin@agroInk.com	admin123 (or any)	#/admin-dashboard	1-Click KYC Approvals, Escrow Ledger Auditing, Assayer Dispatch, Dispute Arbitration
Verified Farmer	Veerappan / Sakthi Vel	farmer@agroInk.com	farmer123	#/farmer-dashboard	Spot Lot Creation, Live Auctions, e-NWR Warehouse Deposits, 75% LTV Loans
Wholesale Buyer	Maran S / Global Agro	buyer@agroInk.com	buyer123	#/buyer-dashboard	Direct Spot Buying, 100% Escrow Funding, Live Bidding, Pre-Dispatch Assayer Requests
Transporter	Apex Fleet Logistics	transporter@agroInk.com	pass123	#/transporter-dashboard	Multi-Axle Fleet Allocation, Corridor Checkpoint Updates, 6-Digit Delivery OTP
Financier / NBFC	Samunnati Capital	financier@agroInk.com	pass123	#/financier-dashboard	30-Day Credit Underwriting, Collateral Vault Liens, 1-Click Capital Disbursement
Warehouse Operator	Salem Agri Cold Hub	warehouse@agroInk.com	pass123	#/warehouse-dashboard	5,000 MT Storage Auditing, Multi-Chamber Temperature Zones, e-NWR Digital Titles

2. Step-by-Step Assistant Testing Scenarios

SCENARIO 1: Identity KYC Gate & 1-Click Admin Approval

Goal: Test that unverified accounts are prevented from posting trades until Admin verifies their identity documents.

Step 1.1 — Login as Farmer & Attempt Listing:

Role: Farmer (farmer@agroInk.com)

Navigation: Open sidebar ➡ Click **Post Produce** (#/farmer-create-listing).

Action: If unverified, observe the **Verification Required Modal** blocking unauthorized submission.

Action: Click **Submit KYC Documents** and upload an Aadhaar / GST document (Image or PDF).

Step 1.2 — Switch to Platform Admin:

Role: Admin (`admin@agroInk.com`).

Navigation: Go to **Admin Dashboard** → **KYC Verification Queue** (`#/admin-verification`).

Action: Locate the farmer in the pending queue.

Action: Click the  **View Document** button to preview the uploaded PDF/Image in the modal.

Action: Click the green  **Approve** button.

Step 1.3 — Verify Instant Privilege Grant:

Role: Switch back to Farmer (`farmer@agroInk.com`).

Visual Check: Notice the farmer now displays the emerald **Verified Producer** shield badge.

Action: Reopen **Post Produce**; the form is now 100% unlocked for trading.

✓ **Expected Outcome:** Secure role-gated onboarding where only KYC-approved producers can initiate commercial trades.

SCENARIO 2: Spot Produce Listing & 0.50% Escrow Procurement

Goal: Test listing creation and verify transparent 0.50% take-rate fee math (0.25% Buyer + 0.25% Farmer).

Step 2.1 — Farmer Creates a Produce Lot:

Role: Farmer (`farmer@agroInk.com`).

Navigation: Go to **Post Produce** (`#/farmer-create-listing`).

Form Values:

Commodity: `Tomato`

Variety: `Hybrid Shivam`

Grade: `Grade A Premium`

Quantity: `500 kg` | **Unit Price:** `₹42 / kg`

Location: `Attur Farmgate Hub, Salem, Tamil Nadu`

Action: Click **Publish Listing**.

Step 2.2 — Buyer Browses & Places Escrow Order:

Role: Wholesale Buyer (`buyer@agroInk.com`).

Navigation: Go to **Marketplace** (`#/buyer-marketplace`).

Action: Search for `Tomato` or filter by `Tamil Nadu`.

Action: Click on the listing card to view details (`#/buyer-listing-detail`).

Action: Click **Direct Purchase / Buy Now**.

Step 2.3 — Audit Fee Breakdown in Order Modal:

Quantity to Buy: Enter `100 kg`.

Check Numbers:

Commodity Subtotal: `100 kg × ₹42 = ₹4,200`

Buyer Platform Escrow Fee (0.25%): `₹10.50`

Total Buyer Deposit: `₹4,210.50`

Action: Click **Confirm & Deposit to Escrow**.

✓ **Expected Outcome:** Order status moves to `confirmed` with funds securely held in escrow.

SCENARIO 3: Pre-Dispatch Assayer Quality Check & Certification

Goal: Test buyer quality protection where certified assayers inspect moisture and grade before dispatch.

Step 3.1 — Buyer Requests Assayer Inspection:

Role: Wholesale Buyer (`buyer@agroInk.com`).

Navigation: Go to **My Orders** (`#/buyer-orders`) and click on the newly placed tomato order.

Action: Locate the **Quality Inspection Desk** card inside the order summary modal.

Action: Click **Inspect Quality (Pre-Dispatch)**.

Action: Check parameters (**Moisture Content** , **Grade Assaying**) and click **Submit Quality Inspection Request**.

Step 3.2 — Admin Assayer Completes Physical Assay:

Role: Platform Admin (**admin@agroInk.com**)

Navigation: Go to **Dispute & Inspection Desk** (**#/admin-disputes**).

Action: Find the inspection request for Order #ORD-XXXX and click **Dispatch / Complete Assay**.

Fill Assay Report:

Assay Grade: **Grade A Premium**

Moisture Content: **10.8%** (Optimal)

Foreign Matter: **0.3%**

Action: Click **Approve & Dispatch Assay Report to Buyer**.

Step 3.3 — Buyer Views Certified Assay Sheet:

Role: Switch back to Buyer (**buyer@agroInk.com**).

Visual Check: Reopen the order. The inspection button is now replaced by a permanent **Readonly Certified Assay Report** showing all certified parameters.

✓ **Expected Outcome:** Zero quality risk for wholesale buyers through independent pre-dispatch certification.

SCENARIO 4: Real-Time Live Reverse-English Auction Arena

Goal: Test live bidding floor with real-time countdown, reserve price checks, and instant win settlement.

Step 4.1 — Farmer Launches Live Auction Floor:

Role: Farmer (**farmer@agroInk.com**)

Navigation: Go to **Create Auction** (**#/farmer-create-auction**).

Parameters: **Basmati Rice (Pusa 1121)** | **20 Tonnes** | Starting Bid: **₹38 / kg** | Reserve: **₹45 / kg** .

Action: Click **Start Live Auction Floor**.

Step 4.2 — Buyer Enters Arena & Places Competitive Bid:

Role: Wholesale Buyer (**buyer@agroInk.com**)

Navigation: Go to **Live Auctions** (**#/buyer-live-auctions**) ➡ Click **Enter Auction Arena** (**#/auction-room**).

Action: Place a bid above reserve price (e.g. **₹46 / kg**).

Visual Check:

Live bid log records the bid timestamp and bidder alias.

Status transitions from **Reserve Not Met** to **Reserve Met ✓** in emerald green.

Real-time timer decrements towards auction close.

✓ **Expected Outcome:** Dynamic, competitive price discovery with automated win conversion into an escrow contract.

SCENARIO 5: Multi-Axle Logistics Manifest & Delivery Confirmation OTP

Goal: Test corridor GPS milestone tracking and secure milestone-based escrow fund release.

Step 5.1 — Transporter Updates Corridor Milestones:

Role: Transporter (**transporter@agroInk.com**)

Navigation: Go to **Logistics Corridor Hub** (**#/transporter-dashboard**).

Action: Select the assigned freight lot.

Action: Assign Truck Type: **Multi-Axle 16T Reefer (TN-28-AC-4091)** .

Action: Advance transit status: **Picked Up** ➡ **In Transit** ➡ **Arrived at Hub** .

Step 5.2 — 6-Digit Delivery Confirmation OTP:

Role: At the delivery location, Buyer shares the **6-digit delivery OTP**.

Action: Transporter enters OTP and clicks **Verify & Complete Delivery**.

Step 5.3 — Verify Automatic Escrow Release:

Role: Platform Admin (`admin@agroInk.com`) ➔ Escrow Ledger (`#/admin-escrow`).

Visual Check: Order updates to `Completed / Escrow Released to Farmer` with the exact 0.50% fee retained by AgroInk.

✓ Expected Outcome: 100% fraud-proof settlement; farmer receives payout only after verified delivery.

🔧 SCENARIO 6: Anti-Circumvention Privacy Chat & Smart Evasion Shield

Goal: Test that contact evasion tactics (split phone digits, domain keywords, number words) are blocked in real time.

Step 6.1 — Open Direct Privacy Channel:

Role: Wholesale Buyer (`buyer@agroInk.com`).

Navigation: Open My Orders (`#/buyer-orders`) ➔ Click the  Privacy Chat drawer button.

Action: Select channel: **Veerappan (Farmer) • Direct Channel**.

Step 6.2 — Test Circumvention Attack Vectors:

Try typing the following messages to test the smart shield:

TEST INPUT SCENARIO	MESSAGE CONTENT TYPED	EXPECTED SMART SHIELD MASKING
Standard Phone Number	Call me at 9840123456 for direct deal	Call me at 98*** ***56 [Protected Phone] for direct deal
Email Address	Send invoice to buyer.agro@gmail.com	Send invoice to [Protected Email]
Domain Fragment	Contact me on yahoo or dot com	Contact me on [Protected Domain] or [Protected Domain]
Split Number Evasion (Sliding Window)	Msg 1: 98755 Msg 2: 67890	Sliding Window Intercepts: Masked to 98*** [Protected Fragment] + Security Warning Bot Alert triggered!
Number Words	call me at nine eight four zero...	Words normalized to digits and masked safely.

Step 6.3 — Verify Real-Time Bidirectional Sync:

Role: Login as Farmer in another tab/browser (`farmer@agroInk.com`).

Visual Check: Both parties see the synchronized conversation thread in real time.

✓ Expected Outcome: Zero PII leakage; traders communicate freely while platform revenue is safeguarded.

🔧 SCENARIO 7: e-NWR Warehouse Deposits & 30-Day Working Capital Financing

Goal: Test electronic Negotiable Warehouse Receipts (e-NWR) and 75% LTV commodity-backed credit underwriting.

Step 7.1 — Farmer Deposits Produce into Cold Storage:

Role: Farmer (`farmer@agroInk.com`) ➔ e-NWR Vault (`#/farmer-inventory`).

Action: Click , select `Salem Agri Cold Hub`, `1000 kg Tomato`, `Chamber B4`.

Visual Check: An electronic title `eNWR-2024-XXXX` is minted.

Step 7.2 — Farmer Applies for 75% LTV Working Capital:

Action: Click **Apply for Working Capital** on the receipt card.

Action: Request `₹25,000` (Max ceiling: `75% of ₹42,000 = ₹31,500`) for `30 Days`.

Action: Click **Submit Working Capital Request**.

Step 7.3 — NBFC Financier Approves & Places Collateral Lien:

Role: Financier (`financier@agroInk.com`) ➔ Underwriting Desk (`#/financier-underwriting`).

Action: Click **Review & Underwrite** ➔ Click **Approve & Disburse Funds**.

Visual Check: Open **Collateral Vault** (`#/financier-collateral-vault`). The receipt is tagged with an active **Lien / Pledge Badge**, locking it from unauthorized withdrawal until repaid.

✓ **Expected Outcome:** Immediate working capital liquidity for farmers without predatory middleman loans.

SCENARIO 8: Live Escrow Account Gateway API & Realtime Webhooks

Goal: Test the virtual nodal Escrow Account engine, simulated bank gateway webhooks, and live balance movements in real time.

Step 8.1 — Open the Live Escrow Gateway Console:

Role: Platform Admin (`admin@agrolnk.com`)

Navigation: Go to **Admin Dashboard** (`#/admin-dashboard`) or **Escrow Ledger** (`#/admin-escrow`).

Action: Click the green ⚡ **Live Escrow API** button in the header.

Visual Check: The **Live Escrow Account Gateway & Nodal API** modal opens displaying:

Nodal Partner: `ICICI Bank Ltd. (Institutional Agri-Nodal Branch)`

Trustee Partner: `Vistra / Axis Trustee Services Ltd. (SEBI Reg.)`

Virtual Nodal Account: `992010884210982` • IFSC: `ICIC0000104`

Live Metrics: Locked in Escrow, Settled Payouts, 0.50% AgroLnk Revenue pool.

Step 8.2 — Test Inward Escrow Deposit Simulator:

Navigation: Switch to the ⚡ **Live Interactive Simulator** tab in the modal.

Action: Enter Commodity: `Tomato`, Trade Amount: `₹5,000`, Buyer: `Maran S`, Farmer: `Veerappan`.

Action: Click ⚡ **Trigger Live Inward Deposit**.

Visual Check:

Instant toast confirmation: `✓ Captured ₹5,012.50 into Escrow Vault! UTR: CMSICICI...`

Locked vault balance increments immediately.

Step 8.3 — Test OTP Milestone Payout & Realtime Webhook Bus:

Action: Click 📄 **Verify OTP & Disburse to Farmer**.

Visual Check: Funds shift from **Locked in Escrow** to **Settled Payouts**, and the 0.50% commission pool increments in real time!

Action: Switch to the 📄 **Live Webhook Logs** tab to view the live JSON event bus:

`webhook.escrow.payment_captured`

`webhook.escrow.payout_disbursed`

✓ **Expected Outcome:** Full institutional escrow transparency with real-time auditability and zero settlement leakage.

3. Quick Sanity & Production Readiness Checklist

✓ **Zero Build Errors:** `npm run build` compiles 1,948+ modules cleanly with 0 errors.

✓ **Zero SQL Policy Collisions:** Supabase schema uses idempotent drop/create definitions.

✓ **Live Nodal Escrow API:** Real-time ledger, simulated webhooks, and virtual account routing operational.

✓ **Responsive UI:** Tested across Desktop, Tablet, and Mobile viewports.

✓ **Channel Deduplication:** Chat displays 1 entry per business partner regardless of order volume.

✓ **Accurate Revenue Math:** 0.25% Buyer + 0.25% Farmer = 0.50% Total Platform Take-Rate.